Solar Filament Segmentation & Space Weather AI

Zero-to-Hundred Master Technical Documentation & System Architecture

1. Executive Introduction: What Are We Solving & Why?

Solar filaments (known as prominences when viewed against the dark sky at the solar edge) are massive clouds of dense, ionized gas (plasma) suspended high above the Sun's chromosphere by intense magnetic field loops. When these magnetic structures become unstable, they violently collapse or erupt into **Coronal Mass Ejections (CMEs)**. CMEs send billions of tons of high-energy charged particles toward Earth, causing severe geomagnetic storms capable of destroying satellite electronics, disrupting GPS navigation, endangering astronauts, and knocking out electrical power grids. Automated, real-time detection of solar filaments is essential for **planetary space weather defense**.

2. The Zero-to-Hundred Pipeline Architecture

The diagram below outlines the sequential transformation of raw telescope observations into calibrated space weather alerts:

Pipeline Step	Input & Operation Applied	Scientific Purpose & Output
STEP 1: Raw Image Ingestion	Full-disk H-alpha (656.3 nm) telescope image (2048x2048 px, 16-bit or 8-bit JPEG/FITS).	Captures chromospheric hydrogen absorption where dark filaments absorb light from below.
STEP 2: Solar Disk Detection	Otsu thresholding + Canny edge detection + minimum enclosing circle calculation.	Accurately isolates the Sun's center coordinates (cx, cy) and radius R_disk from outer space.
STEP 3: 7% Limb Edge Rejection	Safe disk mask created with safe_radius = 0.93 * R_disk.	**CRITICAL FIX:** Completely cuts off the extreme bright outer limb cliff to eliminate false boundary rings.
STEP 4: Limb Darkening Flattening	Uniform local spatial box filter: I_bg = uniform_filter(I, size=h//8); I_flat = I / (I_bg + eps).	Flattens the natural optical brightness drop-off from the center of the Sun to its edge.

2. Zero-to-Hundred Pipeline Walkthrough (Continued)

Pipeline Step	Input & Operation Applied	Scientific Purpose & Output
STEP 5: Contrast Enhancement	CLAHE (Contrast Limited Adaptive Histogram Equalization, clip=2.0, grid=8x8).	Amplifies faint dark filament absorption trenches without blowing out quiet solar regions.
STEP 6: Multi-Scale Top-Hat	Morphological Black Top-Hat transforms using disk kernels k in {7, 13, 21, 31}.	Extracts dark structural valleys and curvilinear depressions narrower than kernel sizes.
STEP 7: Frangi Ridge Filtering	Eigenvalue decomposition of Hessian Matrix H(x,y) across scales sigma in {1.0 to 7.5}.	Calculates vesselness probability response Rb and S, highlighting filamentary tubular structures.
STEP 8: Mask2Former AI Model	5-stage CNN Pixel Decoder + 20 Learnable Queries + Masked Cross-Attention Transformer.	**Learned AI Perception:** Distinguishes genuine filaments from round sunspots and background noise.
STEP 9: Hybrid Fusion & Cleanup	Fused_Prob = 0.5 * P_AI + 0.5 * P_Frangi; Connected Components (min_area >= 40 px).	Combines AI semantic reasoning with sub-pixel ridge sharpness and removes tiny noise specks.
STEP 10: Space Weather Extraction	Zhang-Suen skeletonization, length in km, area in km^2 (Scale: 435.0 km/px), centroid, angle.	Exports actionable space weather telemetry to CSV spreadsheets, JSON catalogs, and live Web UI.

3. Deep Learning Architecture: Mask2Former Vision Transformer

Unlike standard segmentation networks (such as basic U-Net) that treat every pixel in isolation, **Mask2Former** is a state-of-the-art **Transformer-based** architecture designed specifically for complex biological and astronomical structures:

Component	Layer Structure & Dimensions	Functional Role in Filament Extraction
Multi-Scale Pixel Decoder	5-stage hierarchical backbone (C1-C5: 32, 64, 128, 256, 128 channels) with top-down FPN.	Extracts multi-resolution spatial features while maintaining a full 512x512 high-resolution mask feature map.
Learnable Filament Queries	20 query vectors in R^(20 x 128) initialized and learned during backpropagation.	Each query specializes in capturing a specific filament instance or cluster across the solar disk.
Masked Cross-Attention	3-layer Transformer Decoder with attention restricted to predicted foreground regions.	Focuses neural attention **strictly within filament channels**, completely ignoring quiet background solar noise.

4. Are We Using Real AI? (AI Reasoning vs Simple Filters)

YES, ABSOLUTELY! Our system is not just a collection of image filters. It implements a true **2.76 Million Parameter Deep Learning Neural Network (Mask2Former)** trained across 50 epochs on 8,199 expert annotations. Here is the fundamental difference between what classical filters do and what our AI model does:

Capability / Dimension	Classical Filters (Frangi / Top-Hat / Preproc)	Mask2Former Deep Learning AI
Mechanism	Fixed, hand-crafted mathematical formulas (derivatives, thresholds, pixel intensity values).	**2.76 Million learnable weights** optimized via gradient descent to recognize complex visual patterns.
Contextual Reasoning	**Zero reasoning.** Looks only at tiny local 3x3 pixel neighborhoods. Treats all dark pixels equally.	**Global Multi-Head Attention.** Evaluates the entire Sun simultaneously, understanding spatial relationships.
Filament vs Sunspot Discrimination	**FAILS:** Confuses dark circular sunspots and magnetic pores with dark filaments.	**SUCCEEDS:** Distinguishes between round sunspots (rejected) and elongated magnetic filaments (accepted).
Adaptability to Solar Cycle Changes	Brittle; fails when telescope exposure, atmospheric seeing, or solar cycle phase shifts.	Robust; generalizes across varying solar noise, limb darkening, and telescope calibrations.
Chirality & Polarity Awareness	Cannot understand magnetic field orientation or filament chirality (Left/Right barb structure).	Learns feature representations aligned with magnetic Polarity Inversion Lines (PILs).

5. Why We Combine AI + Physics (The Hybrid Advantage)

Neither AI alone nor Classical CV alone is optimal for space weather defense. We use a **Hybrid Decision Engine**:
1. **AI Brain (Mask2Former)** provides the *semantic intelligence* (eliminating false positives from sunspots and limb artifacts).
2. **Classical Physics (Frangi)** provides *continuous edge sharpness* (connecting thin, faint filament spines at sub-pixel resolution).
The combined probability **P_final = 0.50 * P_AI + 0.50 * P_Frangi** achieves a benchmark **69.90% Validation Dice score**.

6. Astronomical Scale & Real-World Space Weather Physics

A segmentation mask in pixel coordinates is useless to solar physicists without physical units. Our system converts raw pixel coordinates into real astrophysical measurements using the GONG solar telescope plate scale:

Space Weather Parameter	Conversion Formula & Mathematical Calibration	Planetary Defense Application
Plate Scale Calibration	1 arcsecond = 725.0 km on the Sun; GONG scale = 0.6 arcsec/px. **KM_PER_PX = 0.6 * 725.0 = 435.0 km/pixel**	Converts 2D telescope pixel coordinates into physical kilometers.
Filament True Length	**L_km = N_skeleton_pixels * 435.0 km**	Filaments exceeding 100,000 km represent critical eruption risks.
Filament Surface Area	**A_km2 = N_mask_pixels * (435.0 km)^2 = N_px * 189,225 km^2**	Estimates total trapped plasma mass available to fuel a CME.
Orientation Angle	**theta = 0.5 * arctan2(2 * mu_11, mu_20 - mu_02)**	Indicates magnetic tilt angle relative to the solar rotational equator.

7. Complete Project Outputs & Deliverables Summary

Deliverable Type	File Path & Technology	Purpose / How to Access
Trained AI Weights	checkpoints/best_model.pth (31.78 MB, PyTorch)	Pre-trained 2.76M Mask2Former weights; loads instantly (0.05s inference).
Interactive Web Dashboard	streamlit_app.py / share.streamlit.io	Live cloud web application for real-time drag-and-drop solar image analysis.
Morphology Spreadsheet	outputs/predictions/*.csv (Excel / Sheets)	Contains ID, Area (km^2), Length (km), BBox, Centroid, and Orientation for every filament.
Space Weather JSON Catalog	outputs/predictions/*.json	Standard structured data format for automated ingestion into space observatory databases.
Visual Overlay Grids	outputs/predictions/*_comparison_grid.png	6-panel visual proof: Original, Preproc, Frangi, AI Heatmap, Mask, and BBox Overlay.